SECTION 1
THE IDEA AND THE TICKET JOURNEY

1.1 SYSTEM FUNCTION.

Private Pass is a test system for event entry. A person can give a copy of a ticket to someone else. Private Pass compares a person's face with the face data in their pass. A prepared gate can do this check without the internet. The design does not keep face data on the server for use across different events.

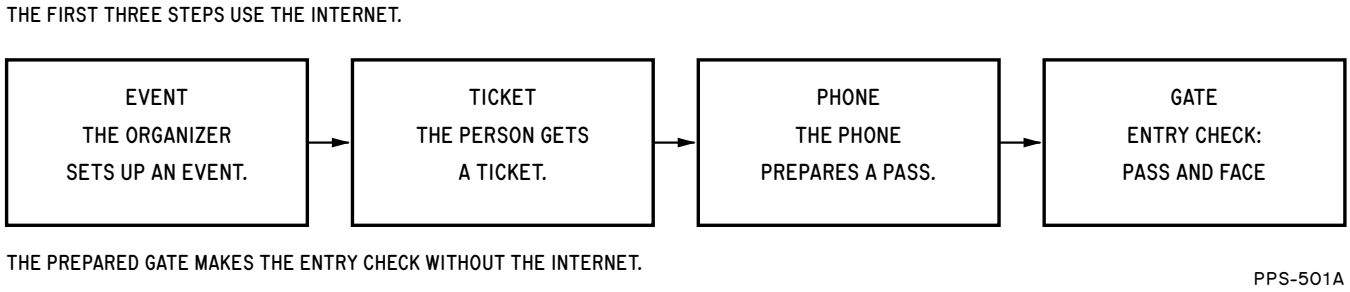


Figure 1. The Ticket Journey

PPS

1.2 THE FACE CHECK.

The phone changes a face image into numbers. These numbers are the face data. The phone changes these numbers for each event. The pass contains encrypted face data. The gate compares this data with the face at the entrance (figure 2).



Figure 2. One Person and One Pass

1.3 PASS PROTECTION.

The gate uses a digital signature to examine the source of a pass. The signature also shows changes to the pass. Encryption changes face data into a form that only a prepared gate can read. The gate must also compare the person's face with the pass.

Table 1. Words Used in This Handout

Term	Meaning
Pass	A digital ticket with encrypted face data.
Gate	The device that does the entry check.
Offline	Without an internet connection.
Server	The computer that keeps ticket records.
Prototype	A test system before use at real events.
Entry record	A saved record of ticket use.

NOTE

The server keeps accounts, events, and ticket records. This design does not remove all privacy risks.

KEY POINT

One person.
One pass.
A local entry check.